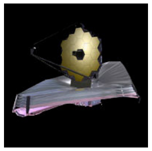

11074 - Resolving Galaxy Evolution: A Public Framework for JWST Slitless Spectroscopy

Cycle: 5, Proposal Category: AR

INVESTIGATORS

Name	***Institution***
Dr. Vicente Estrada-Carpenter (PI)	**Arizona State University**
Dr. Marcin Sawicki (CoI) (CSA Member)	Saint Mary's University
Dr. Casey Papovich (CoI)	Texas A & M University
Prof. Rogier A. Windhorst (CoI)	Arizona State University
Dr. Seth H. Cohen (CoI)	Arizona State University
Mr. Nicholas Foo (CoI)	Arizona State University
Jeremy Favaro (CoI) (CSA Member)	Saint Mary's University, Halifax, Nova Scotia
Dr. Allison Noble (CoI)	Arizona State University

OBSERVATIONS

ABSTRACT

JWST has transformed our view of galaxy assembly, revealing complex internal structures (star-forming clumps, dust gradients, and irregular morphologies) across cosmic time. Yet despite the rapid growth of publicly available slitless spectroscopic data, there is no unified framework to extract and analyze its full spatially resolved potential. We propose to develop and publicly release a pipeline that systematically produces emission-line maps from all available NIRISS and NIRCam Wide Field Slitless Spectroscopy (WFSS) data. We will use an advanced forward-modeling framework that accounts for spatial variations of stellar populations, enabling robust measurements of star formation, metallicity, ionization, and dust attenuation on sub-kiloparsec scales. The resulting database (covering thousands of galaxies from cosmic noon to reionization) will provide the first statistically representative view of how internal structure drives galaxy evolution. By releasing the code, emission line maps, and 1D spectra through MAST, this project will establish a public legacy framework that links JWST imaging and spectroscopy, complements major programs and lays the

foundation for spatially resolved analysis of future Roman grism surveys. This program will transform JWST slitless spectroscopy from a field-by-field tool into a population-level probe of how galaxies build and extinguish their stars across cosmic time.

OBSERVING DESCRIPTION